第 1 章

現状

$$\begin{cases} -\Delta u = \dfrac{1}{\epsilon^2}\bigl(-\alpha u + \beta u^2 - \gamma u^3 - \epsilon(av + bw)\bigr) \\ -\Delta v = u - v \\ -\Delta w = \dfrac{1}{d^2}(u - w) \\ \|u\|_{H_0^1} = N \end{cases} \tag{1.1}$$

$$\begin{cases} -\Delta u = \dfrac{1}{\epsilon^2}\bigl(-\alpha u + \beta u^2 - \gamma u^3 - \epsilon(av + bw)\bigr) \\ -\Delta v = u - v \\ -\Delta w = \dfrac{1}{d^2}(u - w) \\ \sqrt{(\nabla u, \nabla u)_{L^2}} = N \end{cases} \tag{1.2}$$

$$\boldsymbol{F}(\boldsymbol{u}, \epsilon) = K u_h - \frac{-\alpha u_h + \beta u_h^2 - \gamma u_h^3 - \epsilon(aL(K+L)^{-1}L + bL(d^2K+L)^{-1})u_h}{\epsilon^2} \tag{1.3}$$

$$\boldsymbol{\mathcal{F}}(\boldsymbol{X}) = \begin{pmatrix} \boldsymbol{F}(\boldsymbol{u}, \epsilon) \\ \sqrt{\boldsymbol{u}^T K \boldsymbol{u}} - N \end{pmatrix} = \boldsymbol{0} \tag{1.4}$$

$$\begin{pmatrix} \dfrac{\partial \boldsymbol{F}}{\partial \boldsymbol{u}} & \dfrac{\partial \boldsymbol{F}}{\partial \epsilon} \\ \dfrac{\partial g}{\partial \boldsymbol{u}} & \dfrac{\partial g}{\partial \epsilon} \end{pmatrix} \begin{pmatrix} \Delta \boldsymbol{u} \\ \Delta \epsilon \end{pmatrix} = \begin{pmatrix} -\boldsymbol{F}(\boldsymbol{u}, \epsilon) \\ -g(\boldsymbol{u}) \end{pmatrix} \tag{1.5}$$

$$\frac{\partial \boldsymbol{F}}{\partial \boldsymbol{u}} = K - \frac{-\alpha L + 2\beta L u_h - 3\gamma L u_h^2 - \epsilon(aL(K+L)^{-1}L + bL(d^2K+L)^{-1}L)}{\epsilon^2} \tag{1.6}$$

$$\frac{\partial \boldsymbol{F}}{\partial \epsilon} = \frac{2(-\alpha L u_h + \beta u_h^2 - \gamma u_h^3)}{\epsilon^3} - \frac{(aL(K+L)^{-1}L + bL(d^2K+L)^{-1}L)u_h}{\epsilon^2} \tag{1.7}$$

$$\frac{\partial g}{\partial \boldsymbol{u}} = \sqrt{K} \tag{1.8}$$

$$\frac{\partial g}{\partial \epsilon} = 0 \tag{1.9}$$